Week : 11 (9-13 November 2026)

الأسبوع #: من - إلى

Subject/الموضوع: Advanced Physics **Unit Title عنوان الوحدة: Torque** **Grade الصف: 12**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1 **GRASP Launch & Engineering Design** **قانون بويل**	• I can identify the requirements of the rotational equilibrium GRASP and plan a UAE-inspired balanced structure.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	GRASP instructions and rubric on Toddle; physics notebook.
Lesson 2 **Torque Diagrams & Measurements** **Lesson 2** **الحصة الثانية**	• I can construct a labelled scientific diagram showing forces, lever arms, angles and torque directions.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	GRASP instructions and rubric on Toddle; physics notebook.
Lesson 3 **AP Calculations & Proof of Equilibrium** **الحصة الثالثة**	• I can calculate torque using $\tau = rF\sin\theta$ and demonstrate that $\Sigma\tau = 0$.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	GRASP instructions and rubric on Toddle; physics notebook.
Lesson 4 **Scientific Poster & Final Review** **الحصة الرابعة**	• I can communicate my engineering design and support it with labelled diagrams, calculations and a UAE application.	2.4 Support understanding across languages with translations and multilingual tools. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	GRASP instructions and rubric on Toddle; physics notebook.
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	All four lessons take place in class. The physical model is completed at home; the scientific poster, diagrams and calculations are developed in class. Submission details will be shared on Toddle.		

Week #11 الأسبوع#

Technology Integration دمج التكنولوجيا	Digital poster design or presentation software; calculator; Toddle for the GRASP brief and rubric. • • Desmos graphing (θ–t and ω–t). • Physics Toolbox app to estimate ω with video tracking. • Teams submission of torque and angular worksheets.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: trigonometry, moments and equations. Engineering: stability, structures and safety. Design: scientific communication. • Engineering: rotating shafts and wheels in machines. • Design & Technology: gears, pulleys, rotational systems..
External Assessment Integration دمج التقييمات الخارجية	AP Physics-style mathematical reasoning, force diagrams, units and evidence-based justification.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	• Critical thinking through solving equilibrium problems. • Transfer skills: linking linear and rotational analogies. • • Research Skills: • Investigating moments of inertia for real objects. • Using simulations and analyzing data. • • Communication Skills: • Explaining diagrams, torque direction, and reasoning. • Collaborative problem-solving dialogue in groups. • Wellbeing: • Promotes confidence through hands-on activities and manipulatives. • Reduces anxiety by chunking complex rotational concepts into simpler steps. • Encourages peer support in groups.
Standards المعايير	AP Physics 1 – Unit 5: Torque and Rotational Dynamics • Topic 5.1 – Rotational Kinematics: Describe rotational motion using angular quantities. • Topic 5.2 – Connecting Linear and Rotational Motion: Relate force, distance and angle to rotational effects. • Topic 5.3 – Torque: Calculate torque and determine its clockwise or counterclockwise direction. • Topic 5.5 – Rotational Equilibrium: Apply the condition Στ = 0 to analyse balanced systems and justify equilibrium.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can engineers prove that a UAE-inspired structure will remain balanced and safe?

Success Criteria معايير النجاح (Measurable)	Lesson 1 – GRASP Launch & Engineering Design MUST (DOK 1–2): Identify the pivot, forces, distances and GRASP requirements for a balanced structure. SHOULD (DOK 3): Justify the choice of a UAE-inspired design based on safety and feasibility. COULD (DOK 4): Evaluate alternative design ideas and defend the most stable and practical option. Lesson 2 – Torque Diagrams & Measurements MUST (DOK 1–2): Draw a scientific diagram labelling the pivot, at least two forces, distances, angles and torque directions. SHOULD (DOK 3): Analyse clockwise and counterclockwise torque in the proposed model. COULD (DOK 4): Refine a design that would be unbalanced and justify the adjustments. Lesson 3 – AP Calculations & Proof of Equilibrium MUST (DOK 1–2): Calculate torque using $\tau = rF \sin\theta$, with correct units and directions. SHOULD (DOK 3): Use forces and lever arms to demonstrate mathematically that $\Sigma\tau = 0$. COULD (DOK 4): Evaluate how changes in force, angle or distance affect equilibrium and recommend corrections. Lesson 4 – Scientific Poster & Final Review MUST (DOK 1–2): Complete the poster with a labelled diagram, data table, equations and UAE connection. SHOULD (DOK 3): Explain and justify the equilibrium proof and relevance to safe UAE infrastructure. COULD (DOK 4): Evaluate design limitations and propose evidence-based improvements.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging: responsible engineering design. Global Understanding: safe, sustainable infrastructure.
	Area: Innovation and Infrastructure
	Students connect rotational equilibrium to UAE cranes, traffic signs, stadium lighting, solar panel arms and other engineered structures. Their scientific posters explain how balanced designs contribute to public safety, innovation and national development. This topic directly supports the **UAE National Identity Program: Science, Technology, and Engineering (Innovation & Infrastructure domain)** by developing students' ability to apply physics principles—such as torque and rotational equilibrium—to real-world engineering systems. Understanding how forces create balance and stability underpins safe infrastructure design, mechanical efficiency, and problem-solving skills essential for the UAE's Vision 2031 focus on innovation, sustainable development, and future-ready engineers.